What is the Optimal Amount of Biological Noise?

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References

Abstract

Biological noise, the random variations that arise in the genotypes and phenotypes of organisms, can be broadly split into two groups: reproductive noise, i.e., mutation in genotype between parent and offspring, and developmental noise, i.e., irregularities in how genotype dictates phenotype. By increasing genetic and phenotypic variation within a population, biological noise can increase the likelihood of advantageous mutations and traits which, in the genetic case, are retained through natural selection. The amount of biological noise experienced by organisms is strongly impacted by their genotype. Previous work by Weinreich predicts the existence of an evolutionary optimum for both types of noise. Our work aims to find this optimum as a function of spatio-temporal details pertaining to the noise, the timescales of environmental change, and population size. We present preliminary results demonstrating the validity of our simulation framework; because the research is ongoing, these findings represent work in progress toward identifying the evolutionary optima for biological noise. To simulate the evolution of a population, we use a stochastic model that takes into account noise levels, genotype, and phenotype of each individual. By fine tuning these three evolutionary variables and running the simulation for numerous trials, we aim to find the two noise optima and a mathematical solution. This solution could be used to further the field's understanding of pathogenic microbes evolving therapy resistance, which could be applied to preemptively prevent illness. Additionally, this model could be modified to answer evolutionary biological questions regarding the counter-intuitive existence of advantageous polygenetic traits in cases where the addition of only a subset of the mutated genes would be disadvantageous.

First Iteration

Key assumptions of our starting model include fixed population size, non-overlapping generations, all alleles being equally beneficial, and haploid individuals in Wright-Fisher trials. Simulations begin with an initial vector of allele counts, K, where the total population size is fixed at $N = \sum_i K_i$

The next generation is sampled from a multinomial distribution while maintaining constant pop. size where p = K/N:

$$\mathbf{K}_{t+1} \sim \text{Multinomial}(N, \mathbf{p})$$

Simulations continue until one allele reaches fixation ($K_i = N$)

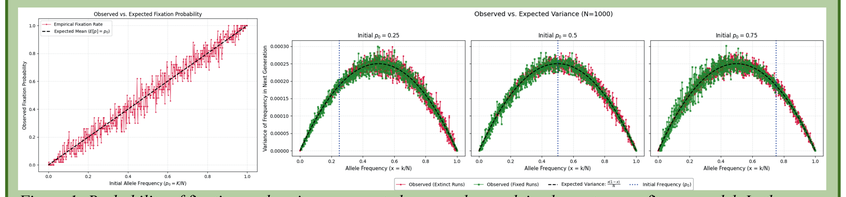


Figure 1: Probability of fixation and variance, expected versus observed, in elementary no-fitness model. In the absense of fitness-based selection, the probability of fixation should be $P_{\text{fix}}(x_0) = x_0$ where $x_0 = \frac{k}{N}$ *and the variance of the next generation at a given frequency should be* $\text{Var}(x) = \frac{x(1-x)}{N}$ where $x = \frac{k}{N}$ *Our simulation results are consistent with both known equations.*

Modeling with Fitness

Relative fitness is used in constant population size models to measure an individual's proportionate success in relation to others. We implemented relative fitness in the second iteration which was the first model to include fitness. This second iteration enables for multiple alleles, and each allele has a relative fitness value that can vary over time.

At each generation, allele fitness values are evaluated and converted into normalized sampling probabilities where $p_i = \frac{K_i F_i(t)}{\sum_j K_j F_j(t)}$ is the fitness of allele *i* at generation *t*. The multinomial process described earlier is then performed.

To confirm the validity of our model, we simulated scenarios with known predicted mathematical results, then compared our simulated data with the expected values. In the next column are examples of the analysis we conducted on the probability of fixation. While not presented here, we also analyzed time to fixation and found that equations accurately verify our time to fixation simulation results. These analyses were performed on the two allele, time-invariant fitness simulation (meaning that all F(t) functions are constants, i.e., independent of t) because that is the scenario modeled by the equations with known expected results. More complex iterations do not have known mathematical checks, so by analyzing early iterations, we gained confidence in the performance of the future sophisticated ones, such as our model with noise.

Haldane's fitness, $1 - \Psi = e^{-(s+1)\Psi}$, is significant within population genetics. A closed form solution has not appeared widely in the biological literature. Xia (2025) expresses the closed form probability of extinction based on Haldane's Poisson distribution law using Lambert W. We use this closed form solution here for the first time within the study of biological noise to analyze probability of fixation (see Figure 2).

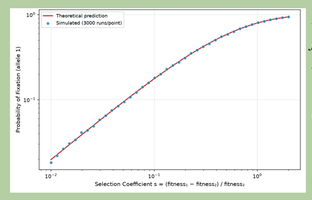


Figure 2: Probability of fixation as a function of selection coefficient, simulated compared to Haldane's fitness. Works for 1 beneficial allele starting in large pop. (5000 here). Our closed form expression $\Psi(s) = \frac{W_0\left(-(s+1)e^{-(s+1)}\right)}{s+1}$ *verifies the accuracy of our two-allele, time invariant fitness model.*

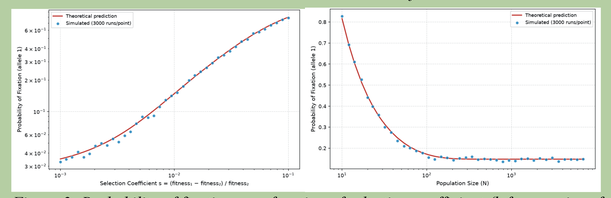


Figure 3: Probability of fixation as a function of selection coefficient (left, pop. size of 300) and population size (right, selection coefficient of 0.01), simulated from two-allele, time invariant fitness model compared to expected $P = \frac{1-e^{-2s}}{1-e^{-2N_e s}}$ *Prediction for introduction of some weakly beneficial allele (8 copies of the observed allele are initially placed in the pop. here). Simulation results align with equation.*

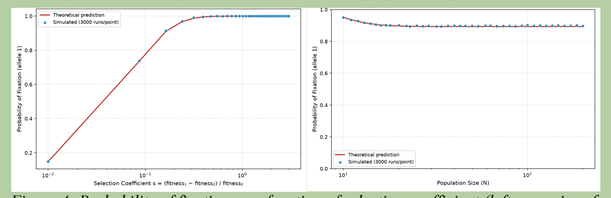


Figure 4: Probability of fixation as a function of selection coefficient (left, pop. size of 300) and population size (right, selection coefficient of 0.15), simulated from two-allele time invariant fitness model compared to Moran model $P = \frac{1-\left(\frac{1}{r}\right)^i}{1-\left(\frac{1}{r}\right)^N}$ *which is uniquely an exact equation, not an estimation, and works for all initial conditions. 8 copies of observed allele are initially present. Simulation results align with equation.*

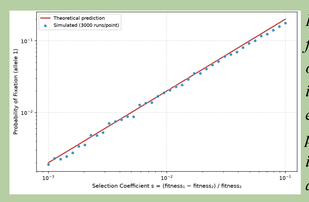


Figure 5: Probability of fixation as a function of selection coefficient (pop. size of 5000), simulated from two-allele, time invariant fitness model compared to expected $P \approx 2s$. *Prediction for large populations with one weakly beneficial initial allele introduced. Simulation results are consistent with known equation.*

Absolute Wrightian Fitness measures the predicted number of offspring of an individual: $W_z = W_{\text{opt}} - (z - z_{\text{opt}})^2$ where W_{opt} represents optimum fitness, meaning the maximum number of offspring any individual could possibly expect to have on average, and z_{opt} represents optimal genotype value. We implemented this fitness model in the third iteration which was the first model with noise. This addition to our model also allowed population size to change.

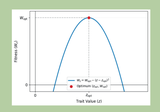


Figure 6: Absolute Wrightian fitness as a function of genotype. Fitness is maximized when genotype equals z_{opt}.

In this third iteration, genotypes are represented as floating point numbers, so there is no set limit to the number of genotypes that can exist within the population. The closer the number representing two genotypes are to one another, the more similar those genotypes, and the traits that they encode, are to one another. Each genotype is assigned a fitness score based on its distance from the optimum.

In our in-progress fourth iteration, genotype determines phenotype, also a float. Phenotype then determines fitness.

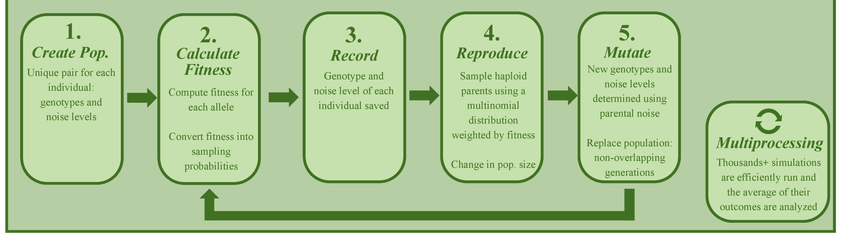
Modeling with Noise

Noise is the introduction of variation and randomness into a system. Genotypic/reproductive noise is caused by mutation. Beginning with our third iteration, we represent the probability of genotypic mutation μ. If the offspring mutates, then the new genotype (z) is deviated using a Gaussian distribution:

$$z_{\text{new}} = z_{\text{parent}} + \Delta z$$
$$\Delta z \sim \mathcal{N}(0, \sigma^2) \quad \text{where} \quad \sigma = \mu_{\text{parent}}$$

Phenotypic/developmental noise describes variation between genotype and phenotype, enabling two individuals with identical genotypes to still have different phenotypes. In our in-progress fourth iteration, the closeness between the float representing phenotype and that representing genotype is selected using a phenotypic noise variable unique to each individual.

Flowchart of Code: Multi-Allele, Time-Invariant Selection Model with Genotypic Noise



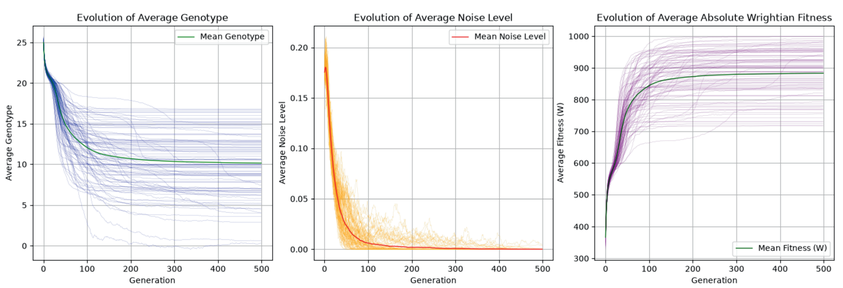


Figure 7: Example compilation of simulation runs for our third iteration, the multi-allele, time-invariant fitness model with genotypic noise. 500 generations were simulated of a population starting at 100 individuals (carrying capacity of 1000 individuals) with random initial genotypes and noise levels (uniformly distributed within bounds of 20 to 30 and 0.05 to 0.3, respectively). The optimal genotype is set to 0 for all time. This was performed 100 times, and the average genotype, noise level, and Wrightian fitness (where Wopt is set to 1000) for all runs were juxtaposed. The average genotype quickly drops towards the optimal genotype, causing the average fitness to rapidly increase. Noise levels briefly increase slightly at the beginning to help find better genotypes (because the original ones are far from optimal), but soon the change in genotype and fitness are slowed then halted by the average noise levels quickly plummeting, resulting in the fixation of alleles. The high Wopt value of 1000 enables genotypes relatively far from the optimum to be beneficial enough to fix, resulting in the long term average fitness values of many functions being lower than the optimum. As expected from the literature, in a non-changing environment (constant optimum genotype), the population evolves towards having no noise because genetic noise's use is introducing new traits that might turn out to be beneficial for a ***new*** *environment.*

Next Steps

- Complete implementation of phenotypic/developmental noise
- Utilize time-variant selection to investigate the influence of temporal and spatial environmental variability
- Locate the developmental and reproductive stable biological noise optima
- Develop a mathematical solution

Continuing to test new iterations' simulation results against proven mathematical estimates relating to probability of fixation, time to fixation, time to loss, selective sweep trajectory time, and variance will enable us to continue to verify accuracy as we further improve our model.